CRITICAL ENERGY FOR DIRECT TRANSIENT STABILITY ASSESSMENT OF A MULTIMACHINE POWER SYSTEM

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ABSTRACT

This paper describes a criterion for determining the critical energy used in direct transient stability assessment of a multimachine power system using the transient energy function method. This criterion provides a method by which the controlling unstable equilibrium point (u.e.p.), for the disturbance under investigation, is identified.

The proposed method is practical and reliable. It is practical in that it offers for a typical multimachine power system a technique for fast determination of the desired controlling u.e.p., among the several candidate ones. It is reliable in that it has been tested on two medium-size power networks, and in rather complex situations. In all cases the correct u.e.p. was identified and verified using detailed system trajectories.

INTRODUCTION

In recent years, considerable progress has been made in the transient stability analysis of multimachine power systems by direct methods [1]. Using transient energy functions, "first swing" transient stability of medium-size power systems has been investigated with a fair degree of success [2-4]. This progress can be largely attributed to two factors: (i) development of functions that adequately describe the transient energy responsible for the separation of one or more generators from the rest of the system, and (ii) better estimate of the critical energy required for these generators to lose synchronism with the system. The latter issue, which is often referred to in the literature as the determination of the "Region of Stability," is of considerable practical significance (an incorrect estimate of that region would result in an erroneous stability assessment), and is the subject dealt with in this paper.

In a previous work by the first author, detailed investigation of the trajectories of generators in the post-disturbance period were made [3]. These investigations verified the findings made by previous investigators, concerning the presence of a controlling unstable equilibrium point (u.e.p.).* The system transient energy evaluated at the controlling u.e.p. is called the critical energy and provides an estimate of the boundary of the region of stability. It was also shown that the controlling u.e.p. could be among a group of u.e.p.'s located in the direction in which the severely disturbed generators move.

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In the course of their investigations the authors of reference [3] made detailed analyses of disturbances caused by faults near a group of seven generators that are located electrically close to each other, in a 17-generator system. Rather complex situations were encountered such that it was difficult to determine the controlling u.e.p. among several candidate u.e.p.'s. This phenomenon was referred to as "the mode of instability." The following modes were encountered:

1. Simple modes, with faults at either end of the area where the seven generators are located. The mode specified by the controlling u.e.p. actually corresponds to that with the angle of the machines initially going unstable being greater than 90°.
2. A mode where the critical energy level is determined by two or more severly disturbed machines, even though only one machine initially loses synchronism. The u.e.p. for this case would have the angles of several machines advanced (i.e., >90°).
3. With faults near the center of the area, a cluster of candidate u.e.p.'s with similar energy level exist. Again only one generator initially loses synchronism, but the u.e.p.'s would have more than one generator with advanced angles.

These results indicated that identifying the controlling u.e.p. is a more complex undertaking than it had been realized. We recall that originally (in the early days of direct stability analysis) it was thought that the critical energy is determined by the u.e.p. "closest" to the post-disturbance equilibrium [5]. Later on it was recognized that the relevant u.e.p. to be considered is that "in the direction" of the disturbed system trajectory [2,6]. Even then it was generally assumed that in the controlling u.e.p. the angles greater than 90° are those of the generators actually losing synchronism, when instability is initially encountered.

The complex modes of instability encountered in the 17-generator system have previously been encountered in the 20-generator system by Athay and co-workers [2] (and later verified by Fouad and Vittal [7]). Furthermore, situations have been encountered (in both systems) where a fault applied near a certain generator results in another generator losing synchronism. These phenomena needed explanation. More significantly, practical direct transient stability assessment requires that the controlling u.e.p. be determined accurately.

A method for determining (for a given disturbance) the controlling u.e.p., among the candidate ones, is presented below.

*A u.e.p. is a solution to the system differential equations $\dot{\underline{x}} = \underline{f}(\underline{x})$, such that $\underline{f}(\underline{x}) = 0$. In power systems, they occur when the velocities are zero and the rotor angle positions of some machines are advanced, i.e. >90°.

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MATHEMATICAL FORMULATION

From Refs. [2,4] the system equations of motion for the classical model in the inertial center frame of reference are given by

$$M_i \dot{\tilde{\omega}}_i = P_i - P_{ei} - \frac{M_i}{M_T} P_{COI} \qquad (1)$$

$$\dot{\theta}_i = \tilde{\omega}_i \qquad i=1,2,\ldots,n$$

where $\tilde{\omega}_i$ and θ_i are the angular velocity and rotor angle position of machine i with respect to the center of inertia respectively.

For generator i,

$$P_i = P_{mi} - E_i^2 G_{ii}$$

$$P_{ei} = \sum_{\substack{j=1\\ j\neq i}}^{n} [C_{ij}\sin(\delta_i-\delta_j) + D_{ij}\cos(\delta_i-\delta_j)]$$

$$C_{ij} = E_i E_j B_{ij}, \qquad D_{ij} = E_i E_j G_{ij}$$

$$P_{COI} = \sum_{i=1}^{n} [P_i - P_{ei}]$$

$$M_T = \sum_{i=1}^{n} M_i$$

M_i = moment of inertia

P_{mi} = mechanical power input

$E_i \underline{/\delta_i}$ = voltage behind transient reactance

G_{ii} = real part of the i^{th} diagonal element of the network's Y-matrix.

G_{ij}, B_{ij} = real and imaginary components of the ij^{th} element of the networks's Y-matrix.

Associated with (1) the energy function V describing the total system transient energy which is always defined for the post-disturbance system is given by [2,3,4]

$$V = 1/2 \sum_{i=1}^{n} M_i \tilde{\omega}_i^2 - \sum_{i=1}^{n} P_i(\theta_i - \theta_i^s) - \sum_{i=1}^{n-1}\sum_{j=i+1}^{n} [C_{ij}(\cos\theta_{ij} - \cos\theta_{ij}^s) - \int_{\theta_i^s+\theta_j^s}^{\theta_i+\theta_j} D_{ij}\cos\theta_{ij}\, d(\theta_i+\theta_j)]. \qquad (2)$$

where $\underline{\theta}^s$ = post-disturbance stable equilibrium point. The first term in (2) represents kinetic energy and the remainder of the terms constitute the potential energy of the system.

In (2) the transfer conductance component of potential energy consists of a path dependent integral. In Ref. [2] by making a linear trajectory approximation the transient energy function evaluated at the u.e.p. is given by

$$V_{cr} = V_{P.E.}^u - \sum_{i=1}^{n} P_i(\theta_i^u - \theta_i^s) - \sum_{i=1}^{n-1}\sum_{j=i+1}^{n} [C_{ij}(\cos\theta_{ij}^u - \cos\theta_{ij}^s) - I_{ij}^u] \qquad (3)$$

where,

$$I_{ij}^u = D_{ij}\frac{\theta_i^u + \theta_j^u - \theta_i^s - \theta_j^s}{\theta_{ij}^u - \theta_{ij}^s}(\sin\theta_{ij}^u - \sin\theta_{ij}^s),$$

where $\underline{\theta}^u$ = unstable equilibrium point.

CRITERION FOR DETERMINING CONTROLLING U.E.P.

When the generators of a multimachine power system are subjected to a disturbance their equilibrium is disturbed. During the ensuing transient, the more severely disturbed generators may or may not lose synchronism with the rest of the system, depending on whether the potential energy absorbing capacity of the network is adequate to convert the transient kinetic energy of the system at the end of the disturbance into potential energy. Therefore, the controlling u.e.p. must properly account for two aspects of the transient phenomena: (i) the effect of the disturbance on the various generators, and (ii) information on the energy absorbing capacity of the post-disturbance network. These two aspects must be incorporated in the criterion to determine the controlling u.e.p.

In a recent work by the authors [8] the energy responsible for separation of one or more generators from the rest (when a large disturbance is applied) was investigated. The normalized energy margin concept (ΔV_n), previously used in [4], was extended to a two machine equivalent formed by a given generator and the rest of the generators. It was found that the relative magnitudes of ΔV_n for the different equivalents give a reliable indication of the <u>degree</u> of <u>stress</u> exerted on different generators by the disturbance. Furthermore, instability occurs when $\Delta V_n = 0$. Using similar reasoning, the following criterion is proposed for determining the controlling u.e.p.

"The post-disturbance trajectory approaches (if the disturbance is large enough) the controlling u.e.p. This is the u.e.p. with lowest normalized potential energy margin at the instant the disturbance is removed."

Stated mathematically, let

$V_{P.E.}$ = potential energy terms in (2)

$V_{K.E.}$ = kinetic energy terms in (2)

Using superscripts u, cl to indicate that the energy is evaluated at the u.e.p. and at fault clearing (or at the end of the disturbance) respectively

$\Delta V_{P.E.}$ = energy absorbing capacity of the post-disturbance network.

$$= V_{P.E.}^u - V_{P.E.}^{cl} \qquad (4)$$

$V_{K.E.}|_{corr}$ = kinetic energy corrected for energy that does not contribute to system separation [3,4].

$$= \frac{1}{2} M_{eq}(\tilde{\omega}_{eq})^2 \quad (5)$$

The normalized potential energy margin $\Delta V_{P.E.}|_n$ is given by

$$\Delta V_{P.E.}|_n = \frac{\Delta V_{P.E.}}{V_{K.E.}|_{corr}} \quad (6)$$

The controlling u.e.p. would be that, among the possible ones in the direction of the trajectory, having the lowest $\Delta V_{P.E.}|_n$.

Practical Determination of the Controlling u.e.p.

Calculating $\Delta V_{P.E.}|_n$ requires knowledge of $\underline{\theta}^u$ for all the candidate u.e.p.'s. Computationally, this can be an expensive and time-consuming task that may make direct stability analysis for large power systems rather impractical. A simplified method is proposed below, that would substantially reduce the computational burden. The method is based on computing $\Delta V_{P.E.}|_n$ for the candiate u.e.p.'s using an approximate value of $\underline{\theta}^u$. However, once the candidate u.e.p. with the lowest value of approx. $\Delta V_{P.E.}|_n$ is determined, its $\underline{\theta}^u$ is computed accurately.

The approximate $\underline{\theta}^u$'s are determined as follows. It can be shown [9,10] that the u.e.p.'s of interest lie in the proximity of the corner points of a polytope. Their approximate positions can be computed from knowledge of the post-disturbance stable equilibrium point $\underline{\theta}^s$. For a given mode, e.g., determined by machines i and j having angles greater than 90°, the approximate $\underline{\theta}^u$ for an n-machine system would be

$$[\underline{\theta}^u_{i,j}]^T \simeq [\theta^s_1, \theta^s_2, \ldots, (\pi-\theta^s_i), \ldots, (\pi-\theta^s_j), \ldots, \theta^s_n] \quad (7)$$

One additional point of practical significance should be made concerning the use of (7) to give an approximate value of $\underline{\theta}^u_{i,j}$, i.e., the effect of the inertial center formulation. In essence it should be noted that the motion of the inertial center of the system from $\underline{\theta}^s$ to $\underline{\theta}^u$ should be accounted for, and the estimated values of $\underline{\theta}^u$ should be adjusted accordingly. The authors' experience indicates that in the vast majority of cases, especially in realistic power systems, this effect can be ignored. However, in small test systems where the inertia of some machines may be disproportionately large, it should be taken into account.*

Procedure for Determining the Controlling u.e.p.

The criterion for determining the controlling u.e.p., by the simplified method (used approximate $\underline{\theta}^u$) is as follows.

Step 1: For the post-disturbance network, the stable equilibrium point $\underline{\theta}^s$ and the admittance matrix Y_{BUS} are determined.

Step 2: The candidate u.e.p.'s for investigation are selected. The rotor angles to be advanced (>90°) are chosen among the machines which are severely disturbed.

Step 3: For each candidate u.e.p. the approximate u.e.p. is determined from (7).

Step 4: Conditions at t_{cl} are determined, including $\underline{\theta}^{cl}$ and $\tilde{\underline{\omega}}^{cl}$.

Step 5: For each candidate u.e.p., $\Delta V_{P.E.}$, $V_{K.E.}|_{corr}$ and approx. $\Delta V_{P.E.}|_n$ is computed using the approx. $\underline{\theta}^u$, $\underline{\theta}^{cl}$ and $\tilde{\underline{\omega}}^{cl}$. The controlling u.e.p. is the one having the lowest approx. $\Delta V_{P.E.}|_n \triangleq M$ ratio.

Step 6: Having determined the mode of instability corresponding to the controlling u.e.p., a suitable minimization technique, (e.g., the Davidon-Fletcher-Powell) could be used to accurately determine the rotor angle positions at that u.e.p.

Examples

The criterion for determining the controlling u.e.p. was applied to the following two medium-size systems:

1. The 17-generator, 163-bus system obtained from the network of the state of Iowa. This system, and the initial operating condition was described in [4]. Three-phase faults on the 345-kV network in the western part of the state were investigated. Seven generators are located in that part of the network which are shown in Figure 1. For faults in that area, these generators tend to be severely disturbed. The faults were selected such that complex modes of instability are encountered, including a case where the generator losing synchronism is different from that where the fault is applied.

2. The 20-generator, 118-bus IEEE system described in [2]. Three-phase faults near generators No. 2 and No. 3 were investigated. The area of the network where these two generators are located is shown in Figure 3. Again, the cases studied include a case where the generator losing synchronism is different from that where the fault is applied.

The procedure followed is based on that outlined in the previous section.

1. For a given fault, typical candidate u.e.p.'s were selected, the approximate $\underline{\theta}^u$'s found, and the corresponding normalized $\Delta V_{P.E.}|_n$ computed; the controlling u.e.p. is thus identified and found.

2. The results are checked by: a- computing the exact normalized $\Delta V_{P.E.}|_n$ for the candidate u.e.p.'s, b- obtaining time solution for the critically unstable case, and comparing the angles of the severely disturbed generators with those in $\underline{\theta}^u$, and by checking the energy level.

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*The same problem is encountered when minimization techniques are used to obtain exact $\underline{\theta}^u$. Starting with the value given by (7), the proper $\underline{\theta}^u$ is usually encountered except for the situation cited above.

The results, for the two power networks, are given below.

The 17-Generator System

Seven fault disturbances were investigated at six locations. In one location (at Ft. Calhoun bus), the fault was repeated with initial operating conditions changed so that 200MW were shifted from Generator #4 to Generator #6, changing the mode of instability for that fault.

Computation of the normalized potential energy margin and identification of the controlling u.e.p., and hence, the critical potential energy, for the various disturbances is shown in Table 1.

To verify the validity of the computed u.e.p., the results of $\underline{\theta}^u$ are checked against the actual trajectory of the system in the critically unstable case, for each fault. At the instant where the potential energy is maximum, the generator rotor angles which exceed 90° are noted. A sample of this data is shown in Figure 2. Table 2 shows, for each fault location, these angles together with the corresponding values in $\underline{\theta}^u$. We note that good agreement is achieved between the advanced generators' rotor angles in the time solution and in $\underline{\theta}^u$. We should also point out that the critical energy values listed in Table 1 give accurate transient stability assessment, i.e., the critical clearing times predicted accurately match those obtained by time solution. This data is not presented here since much of it has previously been published [3,4,7].

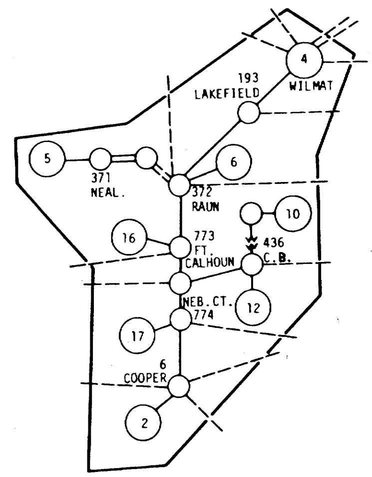


Fig. 1. Area of Interest 17-Generator System

Table 1

Determination of Controlling u.e.p. and $V_{critical}$

17 - Generator System - t_{cl} = 0.15S

Fault Location	Generators With Advanced Angles	$V_{K.E.}\|_{corr}$	$\Delta V_{P.E.}$ Approx $\underline{\theta}^u$	$\Delta V_{P.E.}$ Exact $\underline{\theta}^u$	$\Delta V_{P.E.}\|_n$ Approx $\underline{\theta}^u$	$\Delta V_{P.E.}\|_n$ Exact $\underline{\theta}^u$	Controlling u.e.p. Advanced generators	Controlling u.e.p. V_{cr}
Raun	5,6	6.883	12.689	14.992	1.843	2.1782		
(gen. #6)	5	0.851	8.999	8.859	10.564	10.410	5,6	17.163
	6	6.416	14.437	*	2.250	-		
Council Bluffs #3	10	0.398	2.097	*	5.262	-		
(gen. #12)	12	3.488	8.429	9.2313	2.416	2.646	10,12	12.0249
	10,12	3.840	8.732	10.133	2.274	2.638		
Ft. Calhoun	16	1.067	10.532	10.760	9.871	10.084		
(gen. #16)	10,12,16	2.053	16.354	24.197	7.966	11.786		
	10,12,16,17	2.518	19.885	24.361	7.897	9.674	10,12,16,17	25.1934
	2,10,12,16,17	2.643	21.623	27.184	8.181	10.285		
Cooper	2	3.163	7.480	9.7487	2.365	3.082		
(gen. #2)	2,10,12	3.665	23.182	*	6.325	-	2	11.365
	2,17	4.532	15.678	16.235	3.458	3.583		
Nebraska	17	2.873	10.055	*	3.500	-		
City	16,17	2.143	8.935	*	4.169	-	17	-
(gen. #17)	2,10,12,16,17	4.092	20.871	27.988	5.100	6.8397		
Sub #3455	12	1.691	15.243	*	9.014	-		
	10,12	1.949	16.833	17.545	8.662	9.002	10,12,16,17	26.688
	10,12,16,17	3.096	25.270	24.858	8.162	8.092		
<u>Special Case:</u>	200MW shifted from gen. #4 to gen. #6							
Ft. Calhoun	16	0.940	10.575	10.986	11.246	11.6875		
(gen. #16)	5,6,16	2.098	16.119	19.6235	7.683	9.353	5,6,10, 12,16,17	24.297
	5,6,10,12,16,17	3.573	17.610	23.852	4.929	6.676		

*The minimization program failed to converge for this case.

A point to be noted about the Ft. Calhoun fault is that the critical energy is decided by a u.e.p. corresponding to $\underline{\theta}^u_{10,12,16,17}$ but in the critically cleared case it is only machine #16 which loses synchronism with the rest of the machines. Hence, the controlling u.e.p. plays an important part in deciding the critical energy by indicating the severely disturbed machines but does not indicate the machines which lose synchronism initially.

Another interesting aspect of the Ft. Calhoun case is revealed when 200MW is shifted to Generator #6. The mode of instability changes and the machines initially losing synchronism shifts from Generator #16 to Generators #5,6. The criterion still predicts the correct controlling u.e.p.

Some comment on the Nebraska City fault is in order. Although the DFP (minimization program) did not converge and we do not have a value for the exact $\underline{\theta}^u_{17}$, we estimate $\underline{\theta}^u_{17}$ from the actual system trajectories for the critically unstable and critically stable cases. At the instant where the system potential energy becomes a maximum, the values of $\underline{\theta}$ are shown in Table 3. The estimate of $\underline{\theta}^u$ is also given as the average of the two values.

The results in Table 3 clearly show that the controlling u.e.p. corresponds to $\underline{\theta}^u_{17}$. The value of $\Delta V_{P.E.}|_n$ using the estimated $\underline{\theta}^u$ is 5.37. From Table 1, we note that this value is less than the value obtained for $\underline{\theta}^u_{2,10,12,16,17}$ (6.8397).

Finally, the data in Tables 1,2 and in Figure 2 demonstrate quite clearly the importance of the correct identification of the mode of instability. For the faults at C.B. #3, Ft. Calhoun, and Sub #3455 the controlling u.e.p. include other advanced generator's angles beside the generator which initially loses synchronism. Correct identification of the u.e.p., however, is needed to determine the proper critical value of potential energy required for transient stability assessment.

The Ft. Calhoun case is rather interesting in many ways. We have previously pointed out [3] that there is a "cluster" of u.e.p.'s with similar potential energy levels (in the 25-28 pu range). The u.e.p. with advanced angles 2, 10, 12, 16, 17 was selected by "inspection" of the actual trajectories of the post-disturbance system (we may have also been influenced by the fact that the $\underline{\theta}^u_{2,10,12,16,17}$ was among the easiest to obtain by the minimization program). The transient stability based on that u.e.p. had the greatest error (t_{cr} was in error by about 5%).

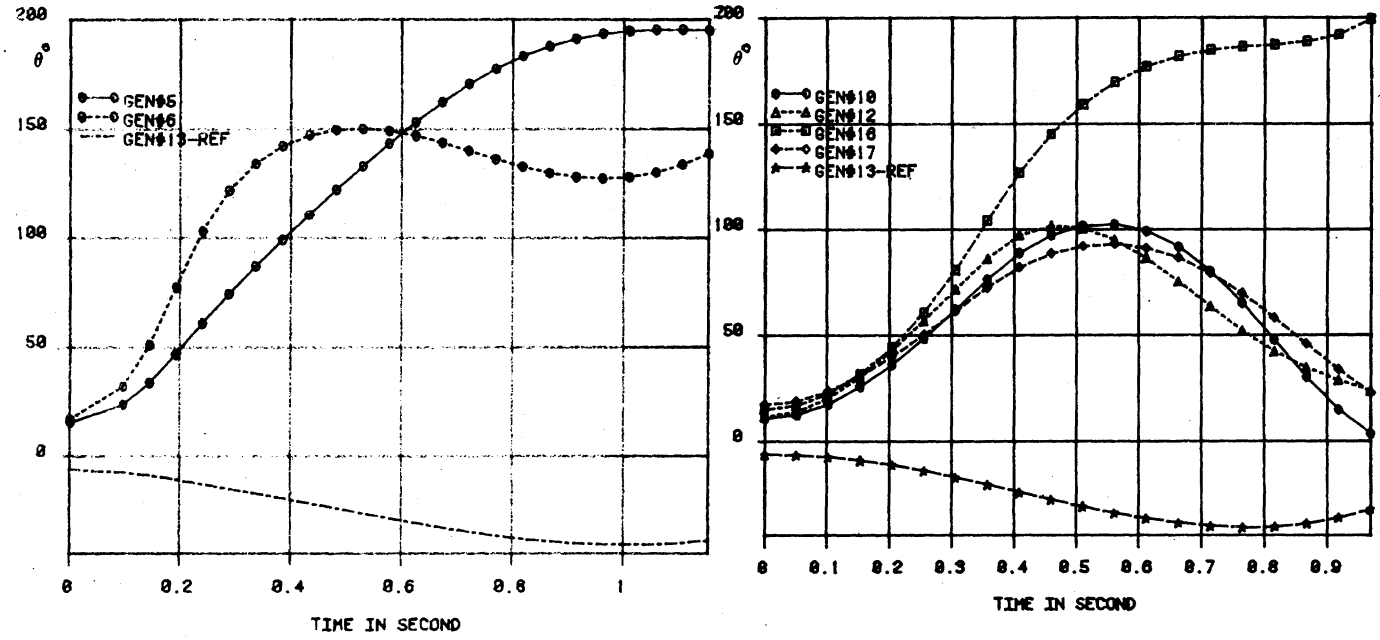
In the present work the criterion for determining the controlling u.e.p. predicted a different u.e.p. Table 1 shows that the correct u.e.p. is $\underline{\theta}^u_{10,12,16,17}$ and a careful check of the system trajectory verifies that this is the correct u.e.p. Furthermore the value of V_{cr} for this u.e.p. gives a much more accurate estimate of the critical clearing time for that fault.

Table 2
17 Generator System
Advanced Generators' Angles at $\underline{\theta}^u$

Fault Location		Advanced Angles $\underline{\theta}^u$	From Critically Unstable Trajectory
Raun	θ_5	163.55	162.24
	θ_6	144.88	143.59
C.B. #3	θ_{10}	154.8	116.49
	θ_{12}	144.7	158.47
Ft. Calhoun	θ_{10}	147.28	102.136
	θ_{12}	164.86	94.747
	θ_{16}	185.29	169.624
	θ_{17}	93.136	92.64
Cooper	θ_2	150.010	151.259
Nebraska Cy	θ_{17}	*	184.044
Sub 3455	θ_{10}	149.12	158.00
	θ_{12}	162.40	196.13
	θ_{16}	173.94	90.44
	θ_{17}	168.875	143.71
Ft.Calhoun	θ_5	153.53	100.91
with 200MW	θ_6	170.866	139.00
shifted to	θ_{10}	122.572	129.39
Gen. 6.	θ_{12}	166.144	138.51
	θ_{16}	182.506	201.53
	θ_{17}	140.378	110.82

Table 3
$\underline{\theta}^u$ At the Instant of $V_{PE/max}$

M/C #	Critically Stable	Critically Unstable	Estimate of $\underline{\theta}^u$
1	-3.7829	5.9109	1.064
2	77.7728	82.4914	80.1321
3	5.75	10.3609	8.055
4	-10.2001	-10.4150	10.307
5	34.9036	39.3193	37.1145
6	38.8038	39.7979	39.300
7	-11.5683	-15.5903	-13.579
8	-6.4687	-8.6928	-7.580
9	-2.1711	-6.4325	-4.302
10	55.8697	46.0684	50.969
11	10.4777	14.0379	12.257
12	50.0054	35.1439	42.57
13	-22.1345	-27.1328	-24.633
14	-22.5168	-27.5113	-25.014
15	-7.0892	-8.3244	-7.7068
16	56.0678	62.9587	59.513
17	145.5127	184.0446	167.778



(a) FAULT AT RAUN.TCL=0.1924 SEC,CRITICALLY UNSTABLE CASE

(b) FAULT AT FT CALHOUN. TCL=0.357 SEC, CRITICALLY UNSTABLE CASE

Fig. 2 17-Generator System Rotor Angles

The 20-Generator System

Two faults were investigated in this system. These were three phase faults located at the terminals of gen. #2 and gen. #3. The faults were cleared without any line switching. Data similar to the ones obtained for the 17-generator system are shown in Tables 4 and 5 and Figure 4.

Data for the $\Delta V_{P.E.}|_n$ for different candidate u.e.p.'s, for the two fault locations, are shown in Table 4. The controlling u.e.p. is identified according to the proposed criterion (Table 4) and verified by inspecting the trajectories of the generators for the critically unstable run (Table 5). Figure 4 shows the trajectories of selected generators.

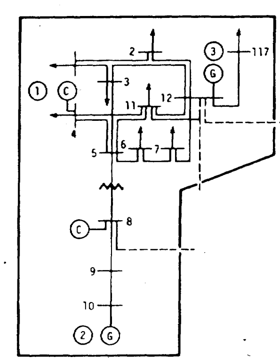


Fig. 3 Area of Special Interest IEEE 118-bus System

Table 4

Determination of Controlling u.e.p. and $V_{critical}$

20 - Generator System - t_{cl} = .15 secs

Fault Location	Generators With Advanced Angles	$V_{K.E.}\|_{corr}$	$\Delta V_{P.E.}$ Approx $\underline{\theta}^u$	$\Delta V_{P.E.}$ Exact $\underline{\theta}^u$	$\Delta V_{P.E.}\|_n$ Approx $\underline{\theta}^u$	$\Delta V_{P.E.}\|_n$ Exact $\underline{\theta}^u$	Controlling u.e.p. Advanced generators	Controlling u.e.p. V_{cr}
Gen. #2	2	1.8756	0.4701	2.798	0.2506	1.491		
	2,3	1.2095	5.009	9.5102	4.14	7.862	2	3.468
	3	0.2785	6.580	*	23.626	-		
Gen. #3	2	0.2797	1.2269	3.295	4.386	11.780		
	2,3	0.4266	5.9109	10.007	13.855	23.45	2	3.468
	3	0.1307	7.1889	*	55.003			

*The minimization program failed to converge for this case.

Table 5

20-Generator System

Advanced Generator Angles at $\underline{\theta}^u$

Fault Location	Advanced Angles	
	$\underline{\theta}^u$	From Critically Unstable Trajectory
Gen. #2	θ_2 125.14	135.74
Gen. #3	θ_2 125.14	118.32

We note that the data presented in Tables 4 and 5 clearly show that the proposed criterion correctly determines the controlling u.e.p., and hence the value of the system critical energy. The case of the fault at generator #3 is of particular interest. While both generators #2 and #3 are severely disturbed, it is generator #2 which initially loses synchronism. The controlling u.e.p. also corresponds to that with θ_2 advanced, which would be difficult to predict without the proposed criterion. The data in Table 5 and the trajectories of Figure 4 confirm these findings.

CONCLUSIONS

1. The controlling u.e.p., and hence V_{cr}, is determined by the weakest link "in the path of the trajectory". This weakest link is identified by lowest $\Delta V_{P.E.}|_n$.

2. The controlling u.e.p. can be correctly identified using computation based on the $\underline{\theta}^u$. Both exact and approximate values of $\Delta V_{P.E.}|_n$ identify the same controlling u.e.p.

3. A procedure for determining V_{cr}, based on approx. $\underline{\theta}^u$, is presented which simplifies the computational burden for large power networks. This is of great practical significance.

4. The procedure has been used to investigate the 17-generator system, and the 20-generator system with excellent results:

    a- Correct mode of instability predicted, and verified by time simulation.

    b- Correct energy level for transient stability assessment is predicted.

5. Procedure will significantly improve prospects of use of direct method.

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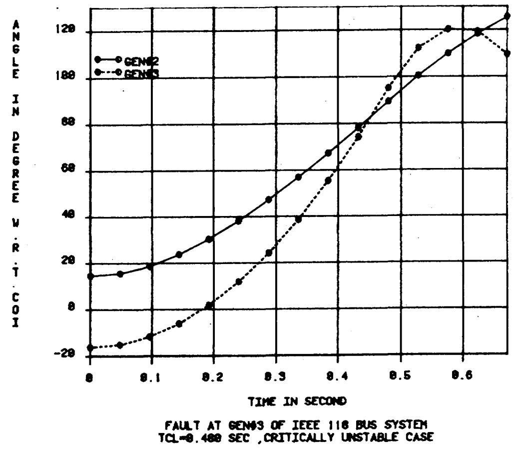

Fig. 4 20-Generator System Rotor Angles

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Discussion

Anjan Bose (Arizona State University, Tempe, AZ): The authors are to be congratulated on devising a simple but effective method of determining the relevant unstable equilibrium point of a multi-machine power system. The efficient identification of the relevant u.e.p., and consequently the mode of instability, makes it feasible to take full advantage of the energy method in determining the stability of the system for a particular disturbance.

It appears that the authors have assumed that the mode of instability is determined soon after the initiation of the fault when all the machines are accelerating. This is, of course, the more common case. However, some of the automatic protective action may be delayed long enough such that some of the machines may have already reached the peaks of their swing curves before the last switchings occur. In such cases, the mode of instability may have to be determined when all the machines are not accelerating. The criterion developed in the paper does not seem to be directly applicable under these conditions and will have to be modified accordingly. The authors comments and the extension of this criterion will be greatly appreciated.

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A. A. Fouad and **V. Vittal:** We wish to thank Dr. Bose for his kind words. His remark concerning the application of the criterion we are proposing is valid and his question is very appropriate.

The proposed criterion for identifying the correct u.e.p. is based on physical reasoning supported by numerous computer simulation results conducted by the authors. The physical reasoning recognizes the network configuration that tends to have the least capacity to "absorb" the excess transient kinetic energy created during the disturbance. The assumption has been, as Dr. Bose correctly points out, that transient stability assessment is made before the critical machines reach the peak of their "swing". Since it is possible that the assessment would be made after the peak of the swing, we should extend the criterion to cover such a situation.

Physical reasoning would lead us to believe that, if stability assessment is made while the critical machines have passed the peak of their swing, the relevant u.e.p. would correspond to that with the highest value of the normalized potential energy margin. Thus the criterion for this situation is the opposite of that for the "normal" case when the assessment is made before the machines reach the peak of their swing. We should also add that we have actually encountered this situation in a study involving generation shedding. In the stable cases, transient stability assessment is made after the critical machines have reached their peak and while their angles are decreasing. The proposed "revised" criterion suggested here seems to give the correct assessment. At this time, however, we have not tested this criterion under many situation. Therefore, these results should be considered preliminary.

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